Cover Letter

To: Guest Editors, IEEE Journal of Selected Topics in Signal Processing
Special Issue: *Autonomous and Evolutive Optimization in Networked AI*
(Lead Guest Editor: Prof. Liang Song; with Profs. J. Liu, A. Dvir, A. Skodras, V. C. M. Leung, and Q. Bi)

Dear Guest Editors,

We are pleased to submit our manuscript, **"Interpretable Federated Kolmogorov–Arnold Surrogates for Autonomous and Evolutive Resource Optimization over Non-Stationary LEO Links,"** for consideration in the Special Issue on *Autonomous and Evolutive Optimization in Networked AI*.

Scope and fit. The Special Issue calls for online, self-evolving mechanisms that let distributed models improve in multi-agent, time-varying environments, with emphasis on the foundations of signal processing for networked AI, networked AI for cognitive communications, and online drift detection and compensation. Our paper addresses exactly this setting. We study autonomous resource optimization over low Earth orbit (LEO) links, where the per-instance optimum has no closed form and the operating regime drifts as satellites pass and interference handovers occur. We cast the problem as learning the *solution map* of an energy-efficiency power-allocation problem, harden it into a mixed-integer program, and decompose it into an inner statistical-learning task and an outer communication-control task whose static optimum we derive in closed form. A federated Kolmogorov–Arnold network learns the solution map online, while a contextual-bandit controller *learns the Lagrangian dual price* of the uplink budget, unifying supervised learning with online reinforcement, as the call envisions.

Contributions and findings. Over twenty seeds with paired significance tests, the KAN surrogate lowers prediction error by about $1.8\times$ and energy-efficiency regret by about $2\times$ versus a parameter-matched multilayer perceptron (Wilcoxon $p < 0.01$), while a linear predictor fails entirely, confirming the map is strongly nonlinear. A learned sparsification controller halves the uplink volume while still outperforming the perceptron, and the KAN extrapolates to unseen interference regimes roughly an order of magnitude more accurately. Uniquely, the additive-spline structure yields closed-form, auditable design rules that a black-box network cannot provide. We compare against a model-based deep-unfolding learn-to-optimize baseline and analyze each component through one-factor ablations.

Relation to prior work. Existing learn-to-optimize methods for wireless resource management are almost exclusively black-box networks (fully connected, deep-unfolded, or graph-based) evaluated under stationary, centralized conditions. Our work differs on three axes that matter for autonomous deployment: it is interpretable (the learned policy reduces to closed-form rules), it is studied under genuine non-stationarity with an online communication budget, and it quantifies sample efficiency and extrapolation rather than only in-distribution accuracy. To our knowledge this is among the first uses of Kolmogorov–Arnold networks as interpretable optimization surrogates inside a networked, federated signal-processing pipeline.

Rigor and transparency. In keeping with reproducible, trustworthy research, all results are multi-seed with standard deviations and Wilcoxon signed-rank tests, every comparison between model classes is parameter-matched and trained under an identical budget, and the ablations toggle one component at a time so that effects are attributable rather than correlational. Importantly, we report transparently the cases where our method does *not* help: online grid self-evolution gives no significant gain over a well-sized static grid, and under abundant centralized data or large channel-estimation error the advantage narrows. The complete code, configurations, and result files that reproduce every number, table, and figure in the paper are openly available.

Declarations. This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved the submission and declare no competing interests. The work was supported in part by the Posts and Telecommunications Institute of Technology (PTIT).

We believe the paper is a strong fit for the Special Issue and hope you find it suitable for peer review. Thank you for your time and consideration.

Sincerely,

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